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Experiment 4: Rule-Based System for Network Troubleshooting

Aim

To develop a Python program that uses a rule-based system to identify common network problems and suggest suitable solutions based on user input.

Algorithm

1. Start the program.
2. Ask the user whether the internet is working, Wi-Fi is connected, and the router is powered on.
3. Collect the user's responses.
4. Apply IF–THEN production rules to identify the possible network problem.
5. Display the identified problem and its suggested solution.
6. Stop the program.

Program

```
print("NETWORK TROUBLESHOOTING SYSTEM")

router = input("Is the router powered on? (yes/no): ").lower()
wifi = input("Is Wi-Fi connected? (yes/no): ").lower()
internet = input("Is the internet working? (yes/no): ").lower()

if router == "no":
    print("Problem: Router is powered off")
    print("Solution: Turn on the router and check the power cable")

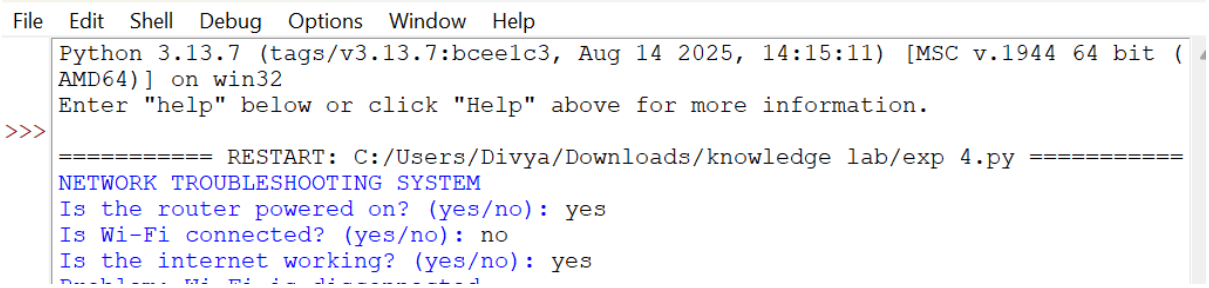
elif wifi == "no":
```

```
    print("Problem: Wi-Fi is disconnected")
    print("Solution: Connect your device to the correct Wi-Fi network")

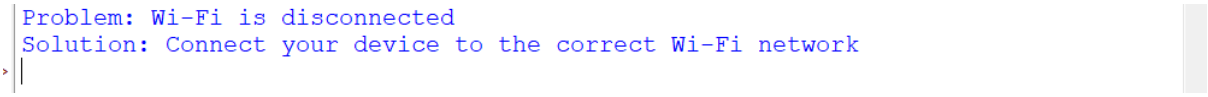
elif internet == "no":
    print("Problem: Internet connection failure")
    print("Solution: Restart the router and contact your ISP if needed")

else:
    print("Problem: No network issue detected")
    print("Solution: Your network is working properly")
```

INPUT:



OUTPUT:



RESULT:

The rule-based network troubleshooting system was implemented successfully using Python.